

EdTech Recorded Lecture Analytics Dashboard

Comprehensive Data Modeling and Performance Analysis

ASSINGMENT POWER BI

Submitted by

KIRTI VARDHAN SINGH

Jan 2026

degree Of

DATA ANALYTICS PROGRAM



1. Executive Summary

This report presents the end-to-end development of an interactive analytics dashboard using a dataset collected from a leading EdTech platform. The primary objective of the project was to transform raw web-scraped data into a well-structured Star Schema data model, enabling efficient analysis and visualization of key business metrics.

The dashboard focuses on analyzing course catalog distribution, learner engagement through viewership, and instructor performance. Built as a three-page interactive solution, it provides stakeholders with meaningful insights into course demand, language accessibility, and the factors associated with high-performing courses. By combining data modeling, transformation, and interactive visualization, the project demonstrates how business intelligence techniques can convert raw data into actionable insights for informed decision-making.

2. Methodology: Data Preparation & Modeling

2.1 Data Cleaning Steps & Profiling

An initial data profiling assessment was conducted on the raw scraped EdTech dataset using Power Query. The original dataset consisted of 8,092 rows and 45 columns. While core fields were populated, several metadata columns required systematic cleaning before analysis:

- **Irrelevant Data:** The auto-generated index field (Column1) was removed to streamline the model.
- **Standardization:** Text-based metric fields were transformed into usable numeric formats. The Views column was cleaned by removing "k", "M", and commas to enable aggregation. The Rating column was stripped of text strings (e.g., "stars") and converted to a decimal format.
- **Feature Engineering:** A binary Subtitle_Available column was created using conditional logic to flag accessibility options. Additionally, inconsistent text-based course durations (months, flexible schedules) were standardized into a continuous numeric Duration_Hours column.

2.2 Relational Data Modeling

To ensure optimal dashboard performance, a Star Schema data model was implemented.

- The main dataset was established as the central fact table (FactCourses).
- Dimension tables (DimCategory, DimLanguage, DimInstructor) were generated by referencing the fact table and removing duplicates.
- To properly analyze skill variety without violating schema integrity, the comma-separated Skills column was normalized using Power Query's advanced split options to create a dedicated DimSkills bridge table.

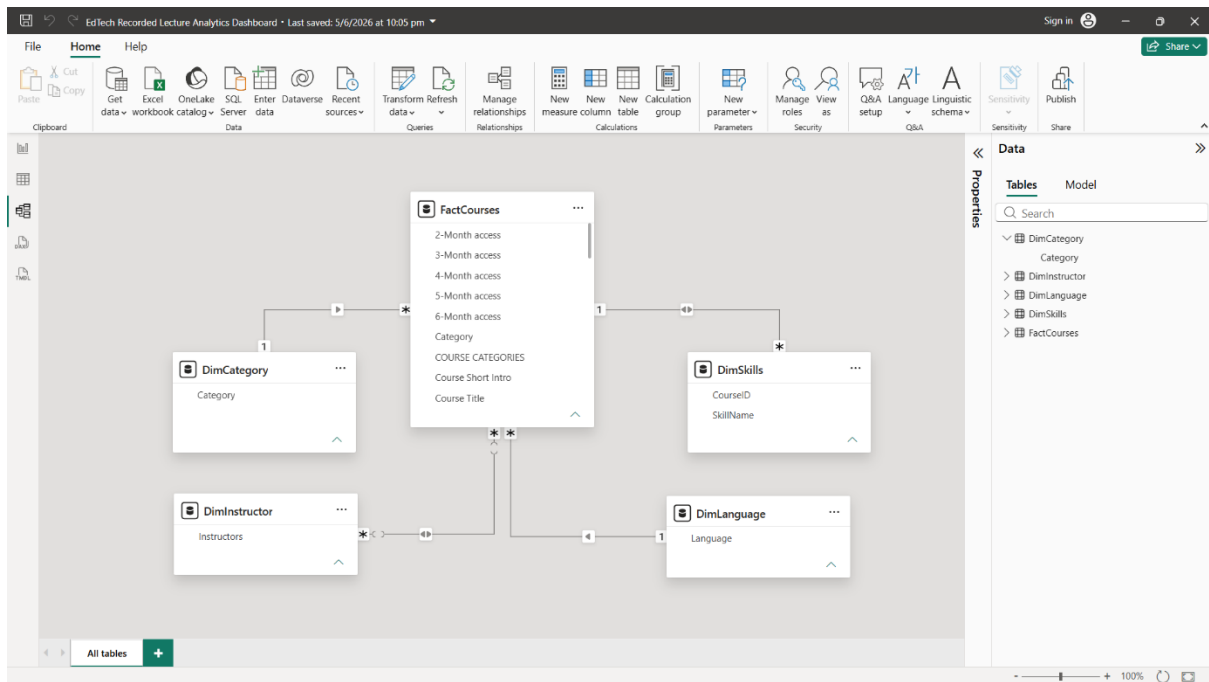


Figure 1: Star Schema / Data Model

3. Dashboard Analysis & Findings

Section 3.1: Catalog Overview

To understand the composition of the platform's offerings, course volume was analyzed across various dimensions.

- **Course Types:** A 100% Stacked Column Chart indicates that standard 'Courses' make up the vast majority of offerings across all categories, while 'Specializations' are notably prominent in technical fields like Data Science.
- **Catalog Hierarchy:** A drill-down matrix utilizing a custom Course Count DAX measure allows stakeholders to navigate from high-level Categories into specific Sub-categories to identify areas with the highest volume of content.
- **Skill Demand:** Leveraging the DimSkills bridge table, a Top 10 clustered bar chart and a hierarchical matrix were developed to instantly identify the most frequently taught skills overall and within specific disciplines.
- **Global Reach:** A Donut Chart highlights the platform's linguistic diversity, providing an immediate understanding of which languages dominate the market share.
- **Leading Technical Skills:** The skill demand visuals reveal that "Data Analysis" and "Python Programming" are the most frequently featured skills across the entire platform, highlighting a massive curriculum emphasis on data literacy.

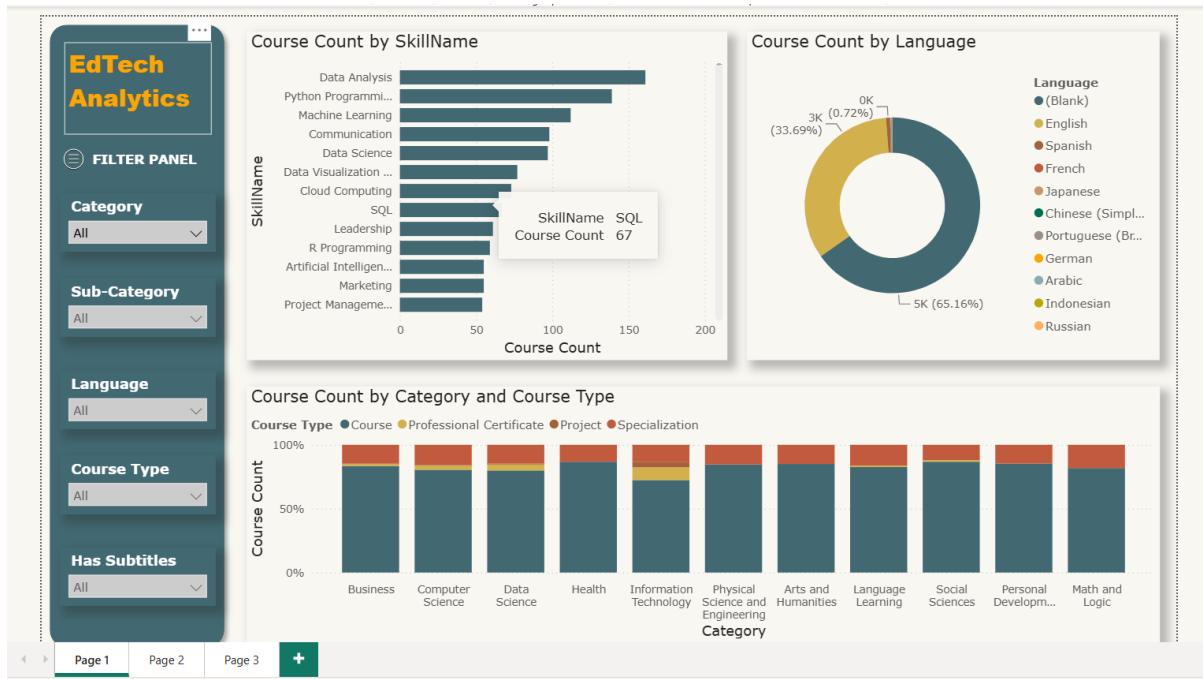


Figure 2: Charts: Course Count Matrix, Top Skills, Language Donut

Section 3.2: Audience Engagement

Viewership metrics were evaluated to identify what drives audience engagement.

- **Category Performance:** Average viewership was calculated using DAX ($AVERAGE(FactCourses[Views_Clean])$) and plotted across Categories, Sub-categories, and Languages to identify high-performing niches.
- **Top Performing Niches:** The visual analysis reveals that highly technical fields, specifically Data Science and Information Technology, generate the highest average viewership, with English and Portuguese being the most consumed languages.
- **Duration vs. Views:** A Scatter Plot correlating course length (X-axis) against average views (Y-axis) was deployed. Visual-level filters were applied to handle statistical outliers, revealing the "sweet spot" for course duration across different colored categories.
- **Accessibility Impact:** A Clustered Column Chart confirms the impact of subtitles on viewership. Paired with interactive slicers, stakeholders can filter this comparison by specific disciplines.
- **The Subtitle Advantage:** The data explicitly demonstrates that courses offering subtitles generate roughly three times the average viewership compared to those without, proving that accessibility features are a major driver of global engagement.

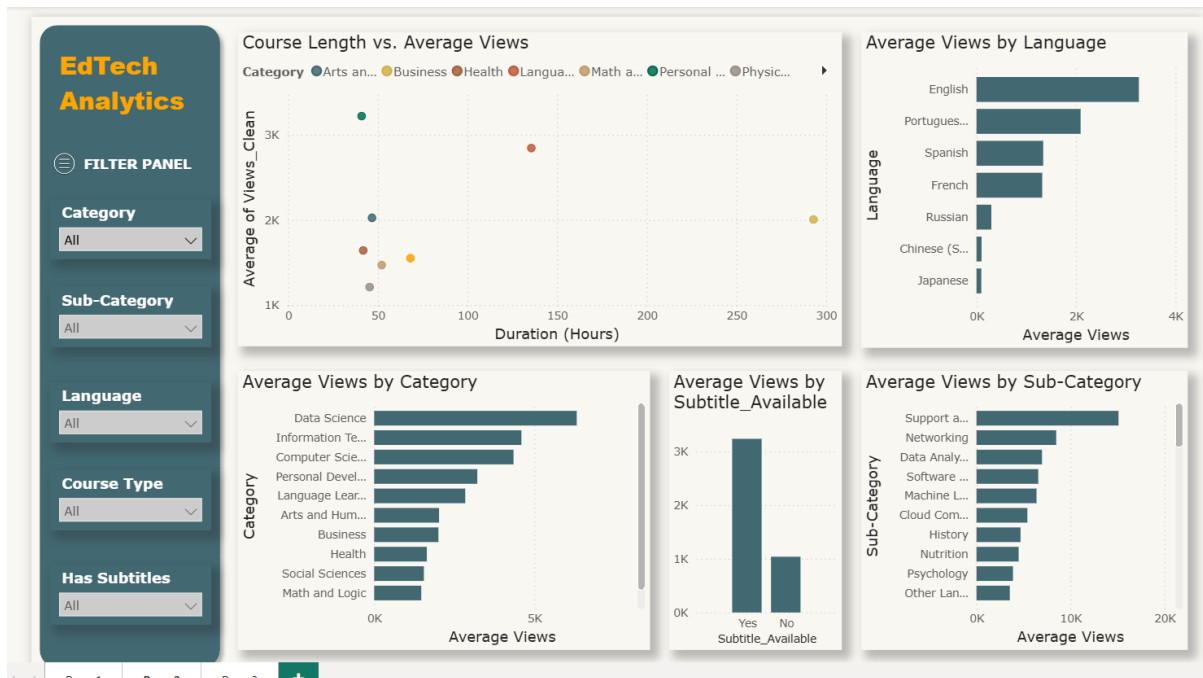


Figure 3: Charts: Avg Views Bar Charts, Duration Scatter Plot, Subtitle Chart

Section 3.3: Advanced Metrics & Instructor Rankings

The final layer of analysis focused on top-tier talent and curriculum depth.

- **Instructor Rankings:** A contextual DAX measure (Instructor Rank) utilizing RANKX and ALL functions was developed to evaluate average ratings dynamically. A static matrix filters this data to showcase only the Top 3 highest-rated instructors per Sub-category.
- **Target Demographics:** A 100% Stacked Column Chart filtered to the "Top 5" categories illustrates the proportional language preferences of the platform's highest-traffic subject areas.
- **Language Dominance in Top Fields:** The demographic breakdown reveals that English accounts for nearly the entirety of total views within the platform's five most popular categories (such as Data Science and Computer Science), indicating a massive market concentration.
- **Skill Variety vs. Viewership:** A final scatter plot correlates a distinct count DAX measure (Skill Variety) against viewership. This allows stakeholders to determine if packing a course with a wider variety of unique skills yields a proportional increase in user engagement.
- **Curriculum Focus vs. Bloat:** The skill variety scatter plot demonstrates that simply increasing the number of unique skills taught does not automatically guarantee higher viewership. In fact, several of the most highly viewed outlier courses feature a highly focused, limited skill set rather than a sprawling curriculum.

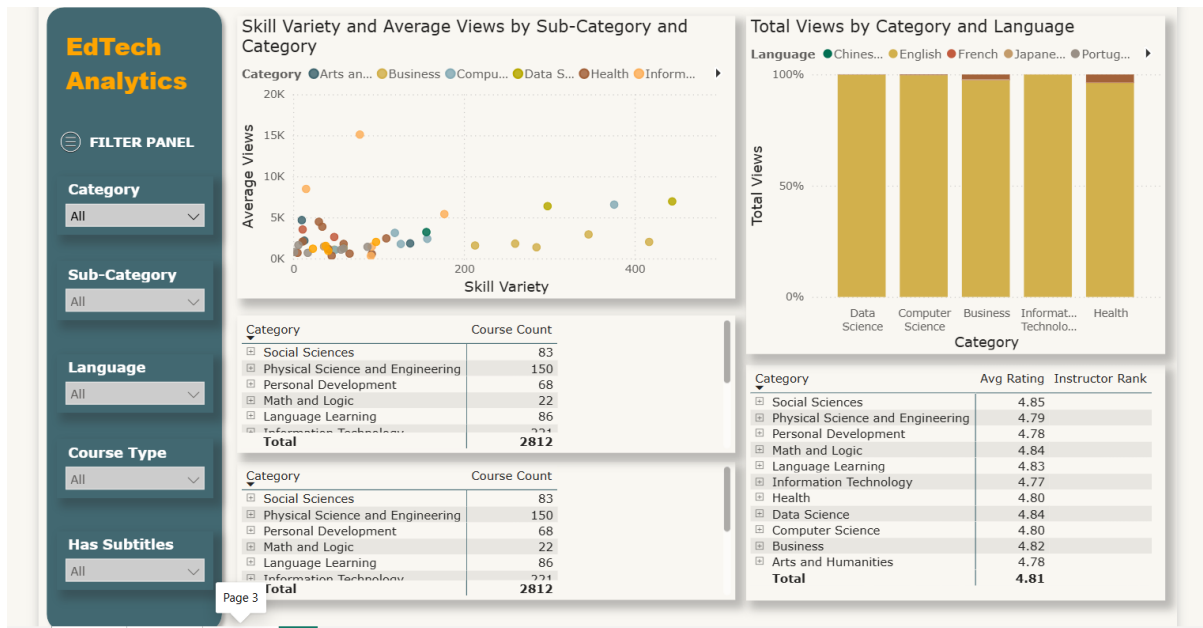


Figure 4: Charts Here: Top 3 Instructors Matrix, Top 5 Language Breakdown, Skill Variety Scatter Plot

4. Key Insights & Recommendations (Category-Wise)

The dashboard analysis reveals several strategic opportunities to optimize platform engagement:

- **Data Science & IT:**
 - **Insight:** These categories exhibit the highest average viewership but also the longest course durations, relying heavily on 'Specializations'.
 - **Recommendation:** Break down marathon Specializations into shorter, targeted "micro-credentials" (under 50 hours) to capture learners who may be intimidated by high time commitments, while maintaining the technical depth that drives high views.
- **Business & Management:**
 - **Insight:** This category has the highest overall course count (volume), creating high competition, but average viewership remains moderate.
 - **Recommendation:** Shift focus from quantity to quality. Leverage the Instructor Rank metric to heavily promote the top 3 highest-rated instructors, and prune outdated or underperforming courses to reduce catalog bloat.
- **Language Learning & Arts:**
 - **Insight:** Viewership heavily favors accessibility; courses with subtitles drastically outperform those without.

- **Recommendation:** Mandate subtitle generation for all future courses in the Arts, Humanities, and Language categories to immediately boost global reach and viewership.

5. Limitations and Future Improvements

While the current data model provides significant insights into catalog performance, there are inherent limitations to the dataset and methodology:

- **Static Data Snapshot (Limitation):** The dataset represents a single web-scraped snapshot in time. It lacks time-series data, meaning we cannot analyze seasonal trends, month-over-month view growth, or shifting skill demands over time.
- **Missing Learner Demographics (Limitation):** The analysis tracks viewership volume but lacks insight into *who* is watching. Without user demographic data (age, geographic location, career level), the ability to target marketing is limited.
- **Future Improvement (Time Intelligence):** Future iterations should integrate an API connection for dynamic data refreshes. Adding a DimDate table would allow for Year-to-Date (YTD) and Quarter-over-Quarter (QoQ) viewership tracking.
- **Future Improvement (Completion Metrics):** Tracking "Views" indicates initial interest, but tracking "Course Completion Rate" would provide a much more accurate metric of true audience engagement and curriculum quality.

GITHUB REPOSITORIES LINK: [EdTech-Analytics-Dashboard](#)